Colouring minimal Cayley graphs: the road from Babai's conjectures to their resolution

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Abstract

In 1978 Babai conjectured that minimal Cayley graphs have a bounded chromatic number. In 1995 he changed his mind and conjectured the opposite.

1 AI/Data usage

I used Claude Opus 5.5 to generate the initial layouts of my latex tikzpictures. Through a prompt like: "I want a circular graph in latex with 3 vertices connected in a circle by 3 edges, two of them are red, one of them is blue"

2 Introduction

3 Preliminaries

3.1 Standard notation for paper

Definition 3.1.1 (Chromatic number)**.** [1] The smallest number of colors required to color the vertices of any graph G, so that any adjacent vertices are never the same color, is called the chromatic number of G. Denoted as $\chi(G)$.

Definition 3.1.2 (Circuit)**.** [1] A *circuit* is a closed walk that can't traverse the same edges twice.

Definition 3.1.3 (Cycle)**.** [1] A *cycle* is a circuit that can't traverse the same vertices twice.

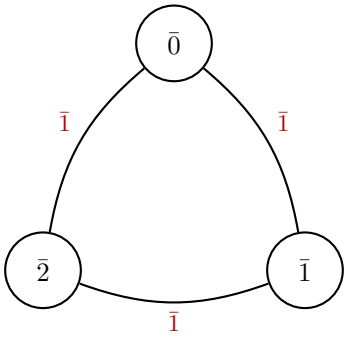
3.2 Cayley graphs with small examples

Definition 3.2.1 (Cayley Graphs)**.** Let G be a group, and let $S \subseteq G$ be a generating set, construct the graph $C(G, S)$ as follows:

- every $g \in G$ is a vertex,
- for every ordered pair (g, s), where $s \cdot g_1 = g_2$ with $s \in S$ and $g_1, g_2 \in G$, set an edge between g_1 and g_2, disallowing identical edges.

The resulting object, $C(G, S)$, is then called a Cayley Graph.

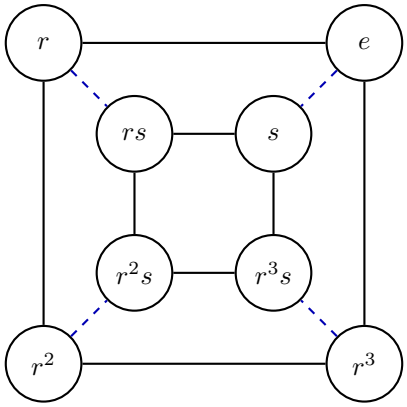
Example.



Let $G = \mathbb{Z}/3\mathbb{Z}$ and $S = \{\bar{1}\} \subseteq G$. Then we have a Cayley graph of this cyclic group.

improve this explanation

Example.



Let $G = D_4$ and $S = \{s, r\} \subseteq G$. Then we have a Cayley graph of this dihedral group.

improve this explanation

3.3 Minimal generating sets

3.4 Hypergraph coloring

4 Babai's question, 1978-1995

4.1 Minimal Cayley graphs are no-lonely-color

Definition 4.1.1 (Quasi-irreducible generating set)**.** Let G be a group, and let $S \subseteq G$ be a generating set. We call S *quasi-irreducible* if the group generated by S equals G, e.g.

$$\langle S \rangle = G,$$

and S can be well-ordered such that no element of S is contained in the group generated by the previous elements.

Example. Let $S = \{2, 3, 4\}$, then $3 - 2 = 1$. Since $\langle 1 \rangle = \mathbb{Z}$, also $\langle S \rangle = \mathbb{Z}$. Now observe that the the ordening $\{2, 4, 3\}$ fails since $4 \in \langle 2 \rangle$, but $\{4, 2, 3\}$ works since $2 \notin \langle 4 \rangle$ and $3 \notin \langle 4, 2 \rangle$. Hence S is a quasi-irreducible generating set of $\mathbb{Z}$.

Counterexample. Let $S = \{1, 2, 3\}$. Notice that

$$\{3, 2, 1\}, \text{ then } 2 \in \langle 1, 3 \rangle,$$
$$\{3, 1, 2\}, \text{ then } 1 \in \langle 2, 3 \rangle,$$
$$\{2, 3, 1\}, \text{ then } 1 \in \langle 2, 3 \rangle,$$
$$\{2, 1, 3\}, \text{ then } 3 \in \langle 1, 2 \rangle,$$
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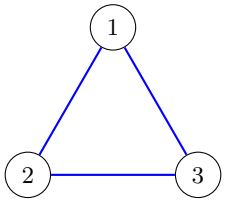
hence S is not a quasi-irreducible generating set of $\mathbb{Z}$.

Definition 4.1.2 (No-pied-circuit graph)**.** A graph G is a *no-pied-circuit* if there exists an edge-coloring such that

- every vertex can only have at most two edges of the same color,
- each circuit contains at least two edges of the same color.

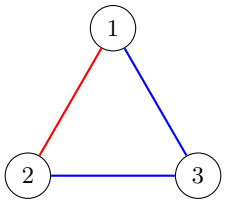
figure out beter definition

Example.



Note that, every vertex has 2 edges of the color blue, and each circuit has 3 edges of the color blue.

Counterexample.



Observe that every circuit only contains the color red once, hence this graph is *not* a no-pied-circuit.

Definition 4.1.3 (No-lonely-color graphs)**.** A graph G is no-lonely-colored if there exists an edge-coloring such that

- every vertex can only have at most two edges of the same color,
- no circuit exists with a color appearing once.

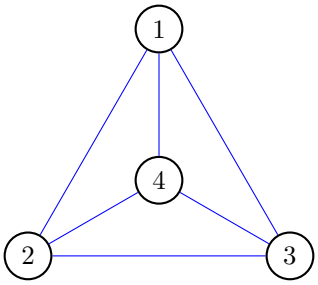
figure out beter definition

Proposition 4.2. *Every no-lonely-color graph is a no-pied-circuit graph.*

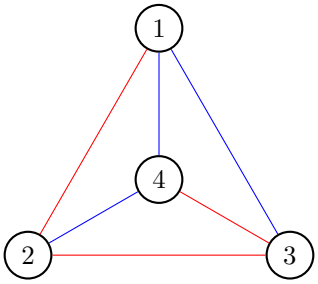
Proof. If no circuit exists with a color appearing once, then every circuit contains at least two edges of the same color. □

Hence the example given for a no-pied-circuit graph (4.1) is also a valid no-lonely-color graph. The converse is not true.

Counterexample.



Note that no circuit existing with a color appearing once means, that all 3-length circuits, which are triangles, are monochromatic. Hence for the complete graph on 4 vertices, K_4, we can only color it in the above depicted way. Which is not a valid no-lonely-color graph since each vertex has 3 edges of the same color connected to it.



Each triangle in this graph satisfies both requirements for a circuit in a no-pied-circuit graph. Furthermore, there are, excluding symmetries, three length 4 circuits on this graph:

$$12341, \quad 12431, \quad 13241,$$

which all have at least two edges of the same color. Hence, this is an example of a graph that is no-pied-circuit but *not* no-lonely-color.

Proposition 4.3. *Replacing circuit with cycle in the definitions of no pied circuit graphs and no-lonely-color graphs gives equivalent definitions.*

make rigorous

Proof. Note that every cycle is already a circuit. So we only have to prove that circuits are the same as cycles in the above definitions. □

finish proof

No pied circuit graphs arise from quasi-irreducible generating sets. Also every no lonely colored graph is a no pied circuit graph. Babai then conjectured that pied-circuits have a bounded chromatic number, which would mean every no lonely colored graph would also have bounded chromatic number.

4.4 Forbidden subgraphs

4.5 Babai's 1978 conjectures

Conjecture 1. *The chromatic number of any no-pied-circuit graph is bounded.*

Conjecture 2.

Both conjecture 2.2 and 3.5 from Babai's 1978 paper

4.6 The 1995 counter-conjecture

5 Evidence in both directions

5.1 Abelian groups are 3-colorable

5.2 More positive results

5.3 Negative results

6 Necessary tools

7 No-lonely-color graphs of large chromatic number

7.1 Tranquil hypergraphs

7.2 Tranquility of C_d

7.3 The construction

7.4 The proof

8 Minimal Cayley graphs of large chromatic number

9 Open questions

10 Appendix

References

[1] Douglas B. West. *Introduction to Graph Theory.* Prentice Hall, 2nd edition, 2001.